EV



OFF-ROAD EV PLATFORM

Powertrain modelling, control and validation on a real vehicle

KICKOFF REVIEW

POL CRUAÑAS · SEPTEMBER 2026

EV



What this project is

An electric mini-quad, used as a bench for power electronics, machines and control. The hardware is finished — the work is modelling it and controlling it.

THE PLANT

The vehicle.
It exists and it runs.

THE DIGITAL TWIN

A Simulink model,
parameterised from measurements.

THE ENGINE

A two-week cycle,
closed by a review with you.

WHAT ALREADY TRANSFERS

- Test benches and measurement campaigns
- Torque and efficiency maps
- Correlating a model against real hardware

WHAT DOES NOT

- Power electronics
- Electric machines in the rotating dq frame
- Embedded control and software architecture

EV The vehicle



Nitro Motors mini-quad: 36 V lead-acid → 48 V lithium.

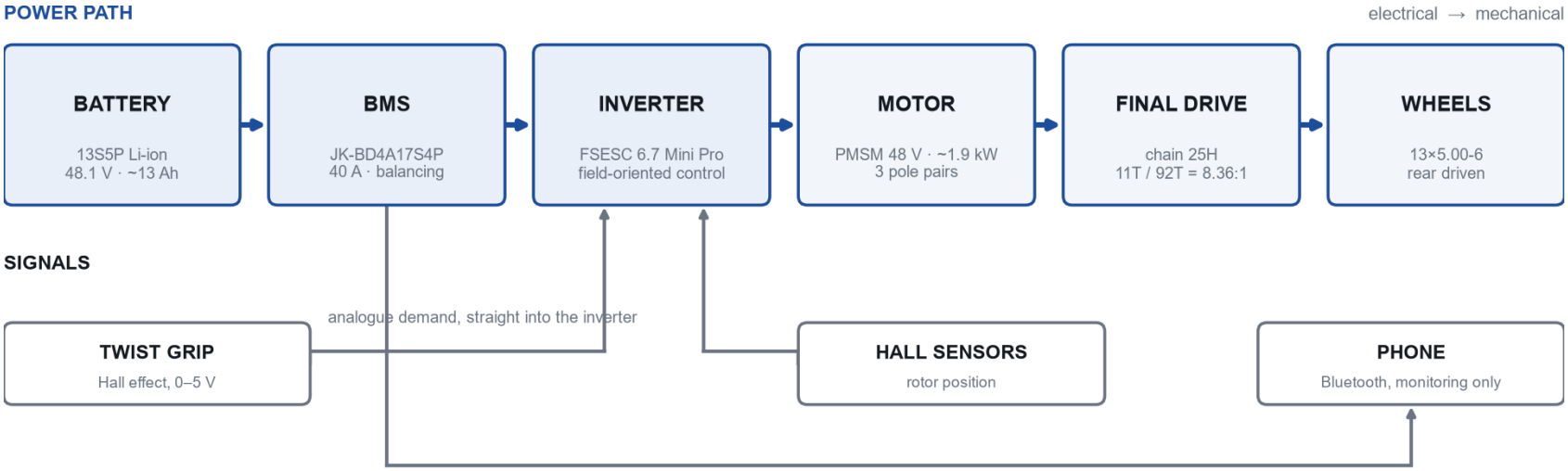
Mass, with driver **≈ 112 kg**	Wheels **13 × 5.00-6 · r ≈ 0.155 m loaded**
Final drive **chain 25H · 11T / 92T = 8.36:1**	Battery **13S5P Li-ion · 48.1 V · ~13 Ah**
Motor **48 V PMSM · ~1.9 kW · 3 pp**	Peak wheel torque **76 N·m at 120 A**

WHY 92 TEETH

		92 T · now	72 T · before
RATIO	$i = z_{crown}/z_{pinion}$	**8.36**	6.55
TRACTION	$F = K_t\, i_{max}\, i / r$	**491 N**	384 N
GRIP LIMIT	$F \le \mu\, m\, g\, x_{rear}$	ASSUMED **536 N**	same
TOP SPEED	$v = \omega_{nl}\, r / i$	**35 km/h**	44 km/h

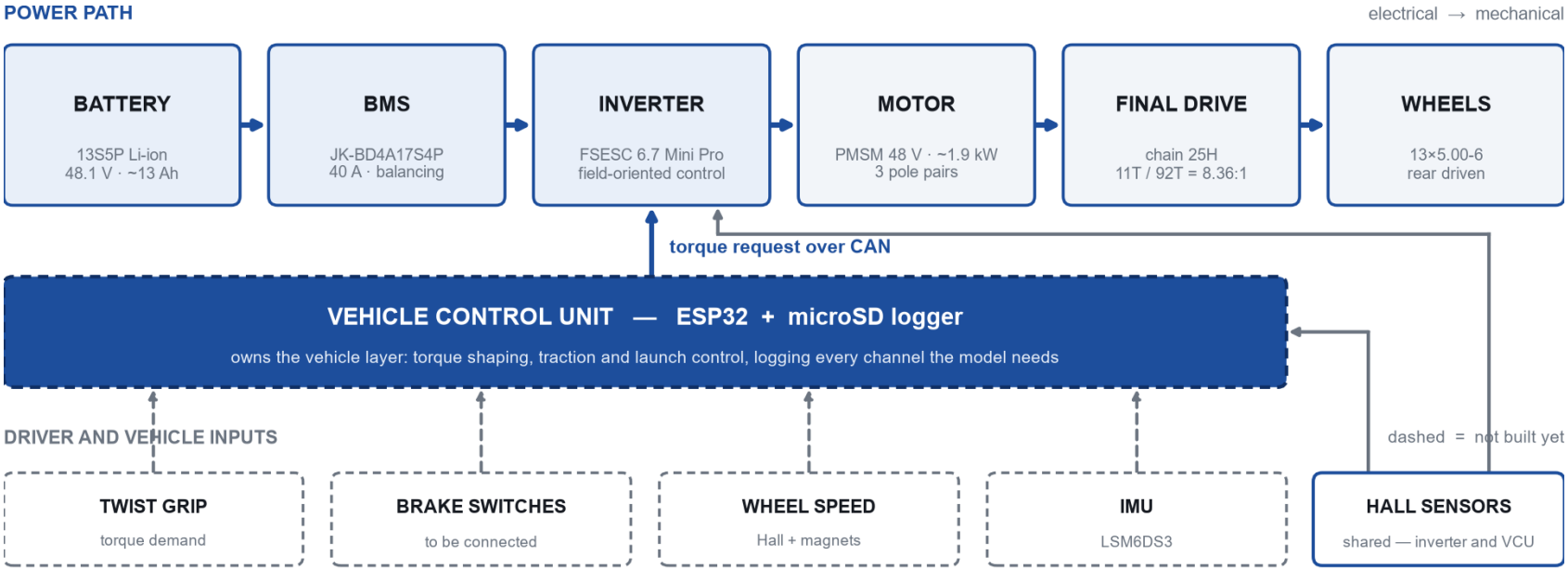
92 T is the largest crown that clears the chassis — and it lands the traction force on the grip limit, so a bigger one would buy nothing. +28 % traction for -22 % top speed. μ, mass and rear weight split are assumed.

System architecture — as it is today



The twist grip drives the inverter directly. Nothing sits between the driver and the torque, and nothing records what happens under load.

System architecture — where it is going



A vehicle control unit takes the driver inputs — brakes included — and sends a torque request over CAN.

The powertrain, component by component

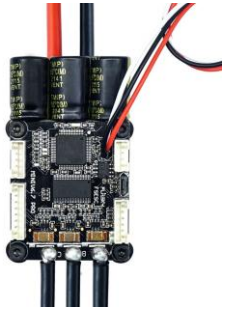


BATTERY

- 13S5P Li-ion · 65 cells
- 48.1 V · ~13 Ah
- JK smart BMS · 40 A

WHY THIS ONE

Configurable and observable: every cell voltage readable from the phone while the vehicle runs.

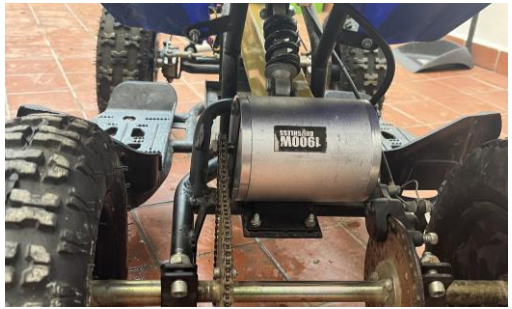


INVERTER

- Flipsky FSESC 6.7 Mini Pro
- 70 A cont. · 200 A peak
- FOC · 30 kHz · CAN

WHY THIS ONE

Open source. I can read the firmware line that computes the torque, and export the whole configuration to XML — which is what parameterises the model.



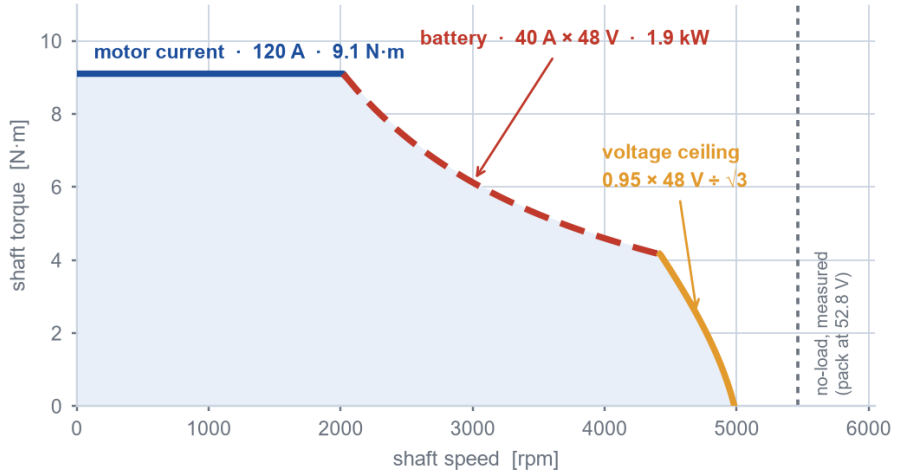
MOTOR

- 48 V PMSM · ~1.9 kW
- 3 pole pairs · Hall
- Urban Fox scooter donor

WHY THIS ONE

Came with the donor, and is not confirmed on paper. It is the component the whole model hangs on, so it had to be measured.

Where the limits actually are



BATTERY

40 A → 1.9 kW — this is the ceiling

MOTOR

120 A → 9.1 N·m — no thermal protection

VOLTAGE

0.95 × 48 V — sets top speed

WHAT THAT MEANS FOR THE MODEL

The inverter, 70 A continuous, never shapes this curve — it limits how long the flat part can be held. The battery gets only what it takes to set the ceiling: no thermal model, not this year.

What the model needs, and where it comes from

TORQUE — needs p, λm

$$T_e = \frac{3}{2} p \lambda_m i_q$$

VOLTAGE, Q AXIS — needs R, Ld, Lq, λm

$$v_q = R\, i_q + L_q \frac{di_q}{dt} + \omega_e (L_d i_d + \lambda_m)$$

MOTION — needs J, T0, b

$$J \frac{d\omega}{dt} = T_e - T_0 - b\,\omega - T_{load}$$

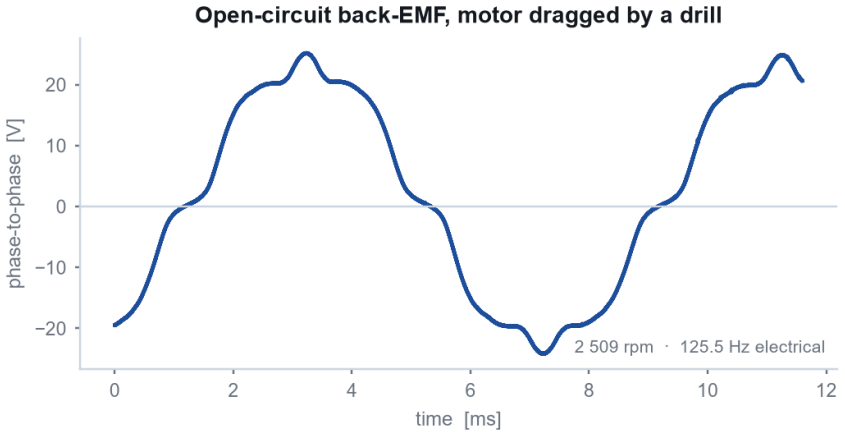
Three equations. Every symbol in them is a number someone has to go and measure.

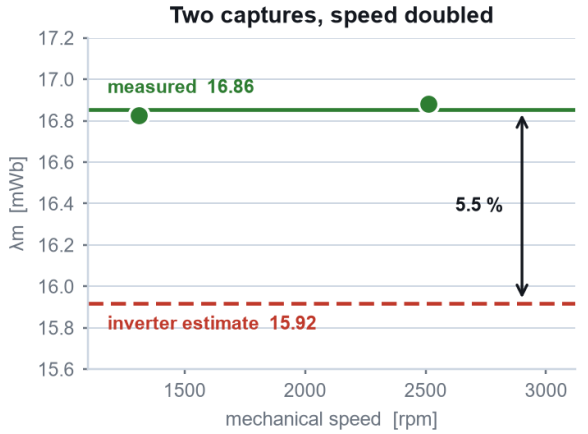
MEASURED ON THIS VEHICLE

p	**3**	Hall pulse count
λm	**16.86 mWb**	back-EMF, scope
Kt	**0.0758 N·m/A**	derived from p and λm
R	**28.0 mΩ**	inverter detection
Ld / Lq	**99 / 113 µH**	inverter detection
J	**1.56·10⁻⁴ kg·m²**	spin-down
T_0 / b	**63 mN·m / 139 µN·m·s/rad**	spin-down

None of this came from a datasheet. The datasheet was wrong.

How the machine was measured





POLE PAIRS

Counted the Hall pulses over one turn of the sprocket: 25 against 25.1 predicted for p = 3, and 33.5 for p = 4.

FLUX LINKAGE

Motor off the inverter, dragged by a drill, scope across two phases. Two speeds agree to 0.17 %.

INERTIA AND DRAG

Three spin-downs anchored on the steady no-load point: J = 1.56·10⁻⁴ kg·m², plus dry and viscous friction.

Next: a coast-down on the vehicle — rolling resistance, drag and vehicle inertia in one test. That is what the straight-line model needs.

EV

How I learn, and what I refuse to delegate

DERIVE IT

A dragged-in FOC block gives a quad that works and an engineer who learned nothing.

- Clarke and Park
- the current loop
- vehicle dynamics

COPY IT

No first principles there, and no shame about it. The budget for deriving goes where the learning is.

- build systems
- protocols
- drivers
- file formats

LET THE AI FILE IT

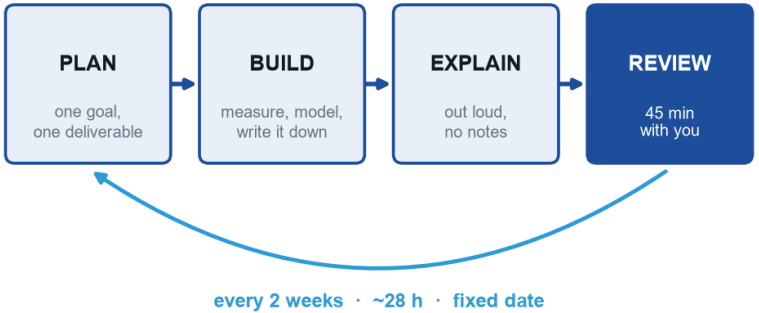
Never the reasoning. An agent that hands me a conclusion I should have reached has negative value.

- notes and plots
- traceability
- formatting
- my own slips

One Git repository holds the measurements, the decisions and the reasons behind them. Nothing important lives only in my head.

How the work is organised

The deadline is the mechanism, not the ritual



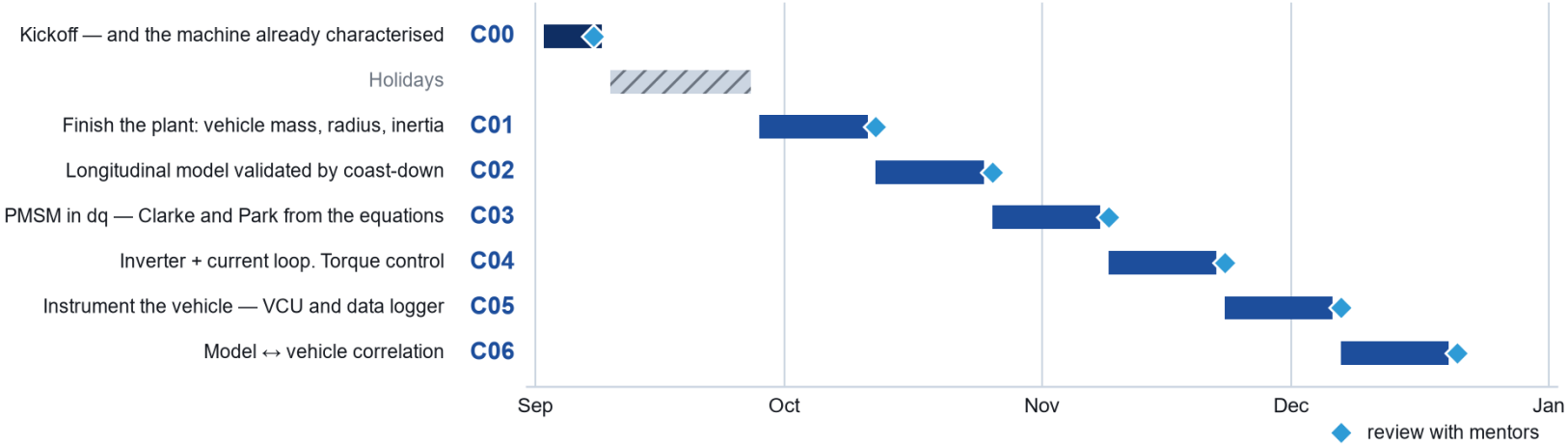
THE 45 MINUTES WITH YOU

5 min	What I said I would do
15 min	What I did, with data
10 min	**What I do not understand**
10 min	**Does the reasoning hold?**
5 min	Goal of the next cycle

The middle twenty minutes are the point. A review where I only show what worked wastes two doctorates.

If I cannot explain it out loud without notes, the cycle is not finished.

The plan to December



AFTER C06 · 2027 Control on the ESP32 · Traction and launch control · MPC on a validated model

End of 2026: a model validated against the real vehicle.

The machine came in ahead of plan, so C01 moves on to the vehicle.

What I need from you

Three things. They are worth more to me than technical answers.

01 **Hold me to a date**

Without an external deadline I postpone indefinitely. The date being fixed is the mechanism — the meeting itself is secondary.

02 **Attack the reasoning, not the result**

Do not ask whether the number is right. Ask why I chose that method, and what would have to be true for it to be wrong.

03 **Tell me when I explain badly**

If an explanation does not land, that is usually not a language problem. It is me not having understood it.

What I am not asking for: that you validate field-oriented control or electric-machine theory. That gap I close elsewhere — you are not being examined on a domain that is not yours.



What I am asking for today

12 Oct

First full review

C01 — closing the vehicle side

Every 2 weeks

Cadence

Monday, 45 minutes, six reviews to December

3 of 3

Machine parameters measured

pole pairs · flux linkage · inertia

Confirm the slot, and I will put all six review dates in the calendar before I leave.